Broad rationale for conducting the research, the objectives, and methods

The ocean's surface layer is crucial for Earth's climate because it absorbs excess heat from the atmosphere, helping to stabilize global temperatures. The sea surface temperature (SST) memory timescale refers to how long anomalies in SST persist before dissipating. Observations show that the persistence of SST anomalies has been increasing, but it remains unclear how external forcings, such as greenhouse gases and anthropogenic aerosols, contribute to this trend. Several processes can influence SST memory, including the amount of heat stored in the mixed layer and the rate at which the ocean or atmosphere dissipates heat via atmospheric or oceanic circulations. For example, a deeper mixed layer stores more heat and dampens short-term fluctuations, affecting how the ocean responds to climate variability. Climate model historical simulations offer opportunity to investigate the past ocean changes. This project will analyze climate model simulations and compare them with observational data to quantify the contributions of different physical processes to changes in ocean thermal inertia. This will help reveal what factors most influence how the ocean stores and releases heat, to gain a clearer understanding of how climate change drives extreme events and affects marine ecosystems, especially as global ocean heat content has reached record highs in recent years.

The goal of this project is to understand how different external forces affect the ocean's thermal inertia and to explore what drives changes in the ocean's heat storage. The project will focus on the role of atmospheric and oceanic factors in shaping the sea surface temperature and on breaking down the energy budget of the mixed layer to see which processes contribute most to thermal inertia. By studying these components, the project aims to clarify how the ocean absorbs, stores, and releases heat, which is important for predicting climate variability and extreme events that impact marine ecosystems.

The methods of the project are to: (1) analyze outputs from different climate models using the *Xarray* package to handle large climate simulations, (2) apply an autoregression model to calculate the ocean's thermal inertia, (3) use energy budget equations of the mixed layer to quantify how different processes contribute to heat storage and release. All data analysis will be done on *NCAR's* supercomputer via *JupyterHub*, which allows full remote access through a web browser.